Image Quality Analysis & Enhancement

Region-based diagnosis and correction using signal processing

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Problem Statement

Localized Degradations in Images

Input Image

Original

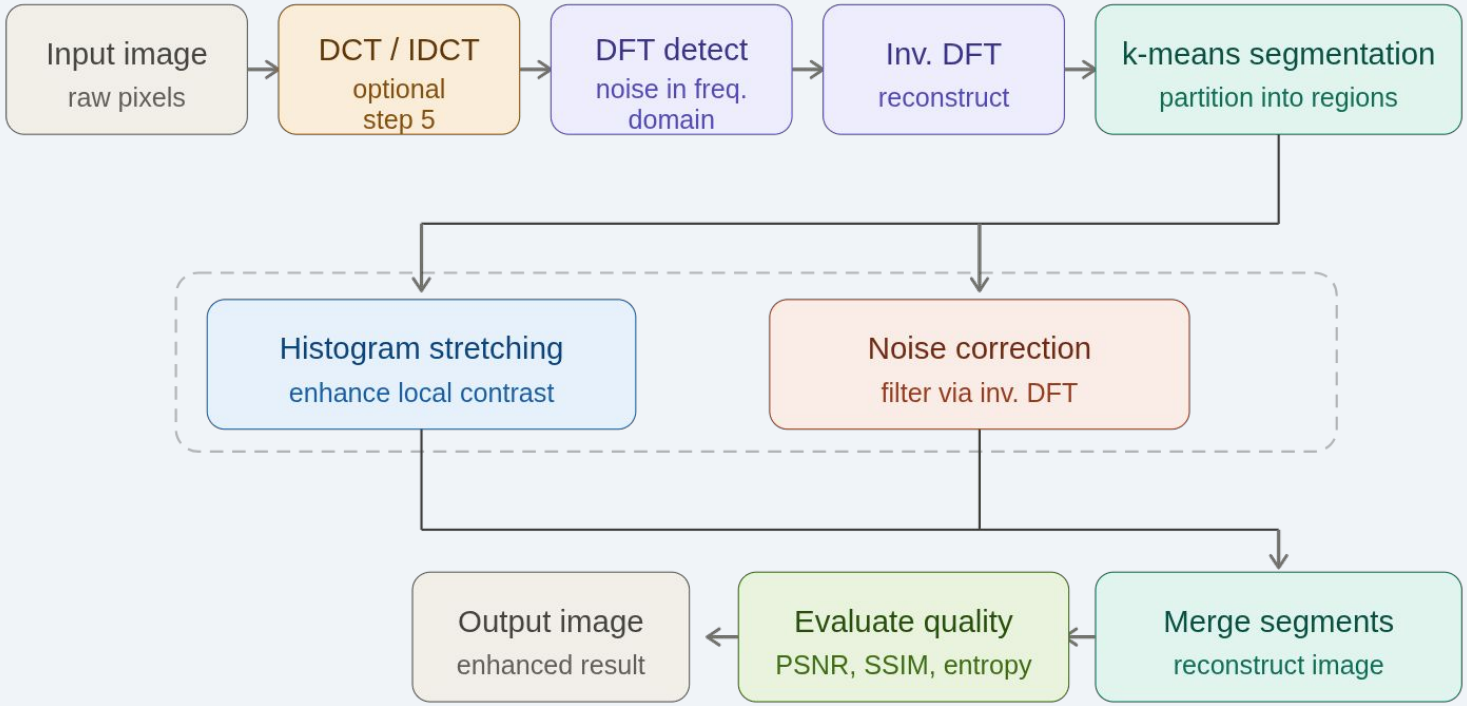
Global Enhancement

Local Enhancement

Why Global Correction on Image Fails

- **Local vs Global:** Global methods apply the same algorithm across the entire image. Different regions need different algorithms.
- **Regional Conflict:** Denoising may blur already blurry regions or wash out fine exposure details, while not addressing local artifacts
- **Specific Region Requirements:**
  - **Noisy sky → frequency domain filtering**
  - **Blurry building → sharpening**
  - **Dark regions → tone correction**
  - **JPEG blocks → DCT-based smoothing**

Solution


Input image
raw pixels
DCT / IDCT
optional
step 5
DFT detect
noise in freq.
domain
Inv. DFT
reconstruct
k-means segmentation
partition into regions
Histogram stretching
enhance local contrast
Noise correction
filter via inv. DFT
Output image
enhanced result
Evaluate quality
PSNR, SSIM, entropy
Merge segments
reconstruct image

Stage 1 + 2 : Discrete Fourier Transform & K-means Segmentation

DFT Frequency Analysis

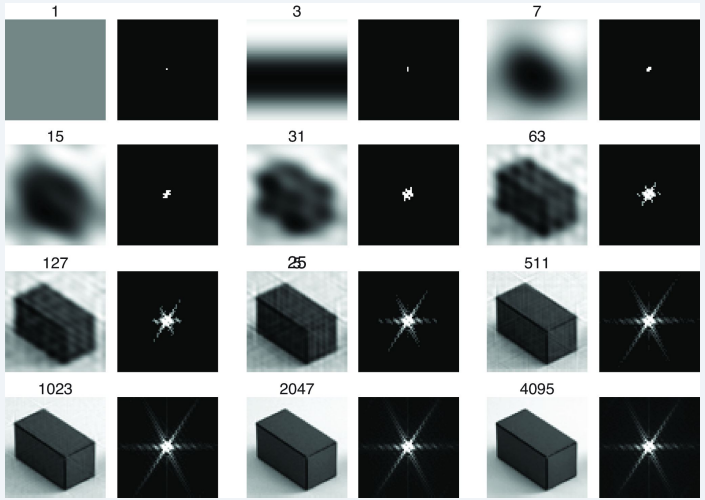
Blur & Noise Detection

$$\text{blurScore} = \frac{\text{highFreqEnergy}}{\text{totalEnergy}}$$

If blurScore $< T_b \Rightarrow$ region is blurry

$$\text{noiseScore} = \text{Var}\left(|F(u,v)|_{\text{high-freq}}\right)$$

If noiseScore $< T_n \Rightarrow$ random noise present



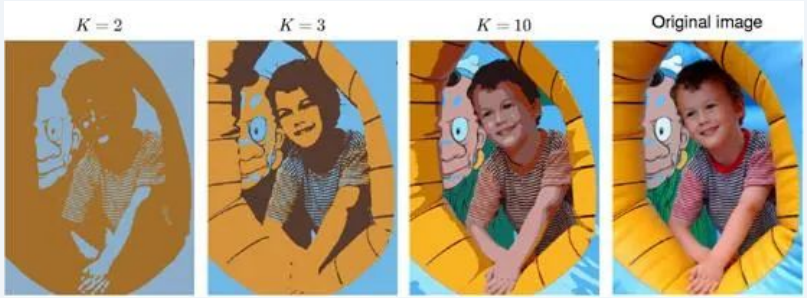
Reconstructing an image from the Fourier coefficients of the largest amplitude.

K-Means Segmentation (LAB color space)

Cluster on L, A, B channels

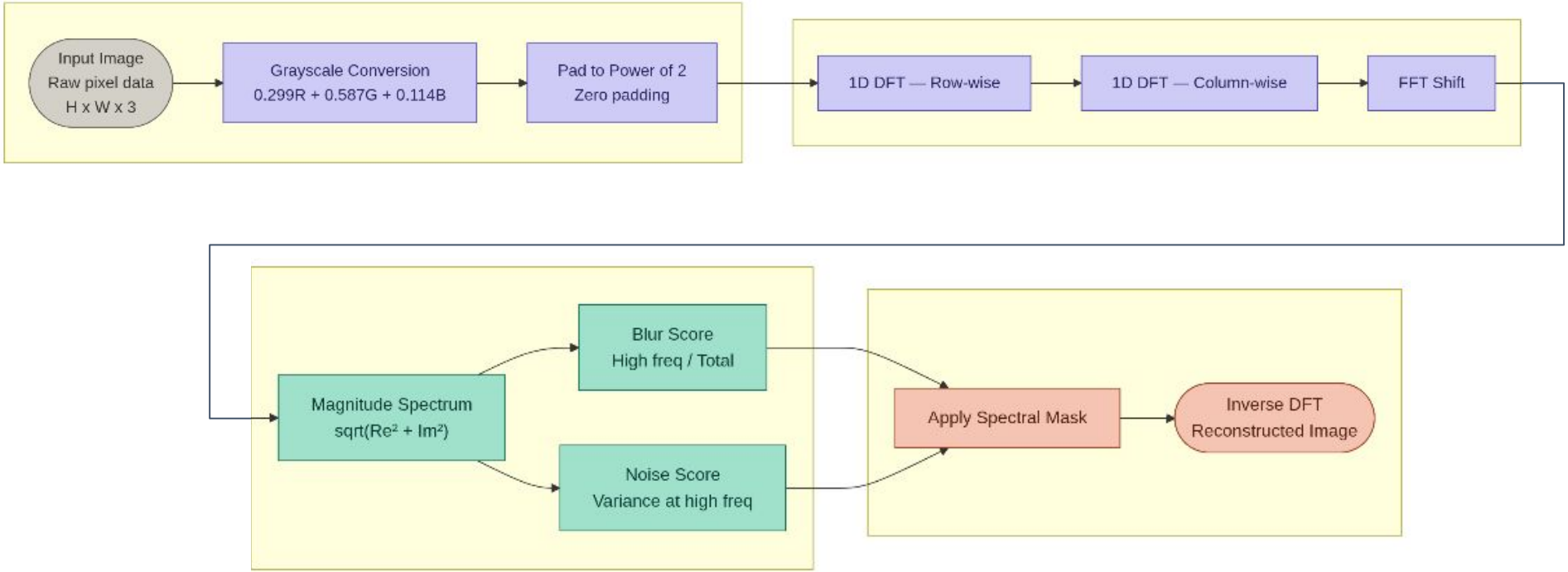
K = 3–4 regions (sky / shadow / midtone / object)

Produces a per-pixel region mask



- **Region-based processing:** for targeted corrections
- **Color clustering:** for object/background separation
- **Image simplification:** for compression & faster processing
- **Preprocessing step:** helps in further tasks like detection and enhancement

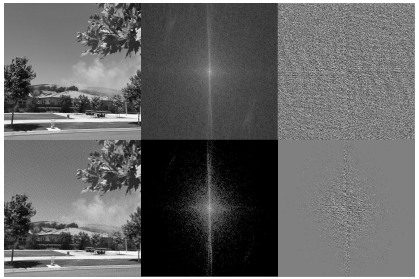
DFT Frequency Analysis



Stage 3 : Per-Region Correction

Sharpening

Boost high-freq components in DFT domain, then apply IDFT to get sharpened region.

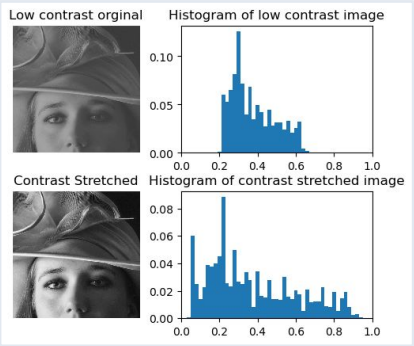


Denoising

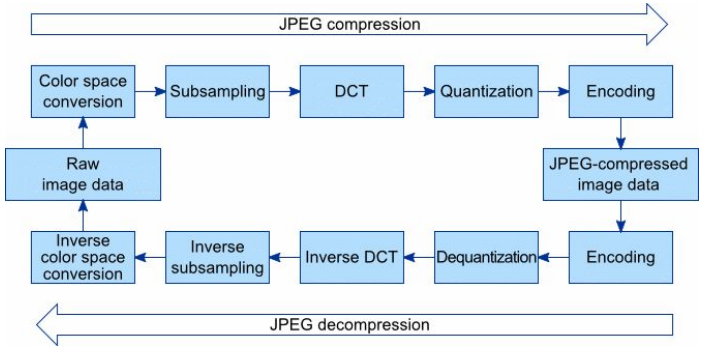
Zero out high-freq components beyond a cutoff radius. Inverse DFT smooths the region.

Exposure → Stretch

Per-region histogram stretching: remap pixel intensities to full [0, 255] range.



Blockiness removal by Inverse DCT



Stage 4 : Evaluation Metrics

PSNR

Peak Signal-to-Noise Ratio

Measures pixel-level fidelity.
Higher => better reconstruction.
Reported in dB.

$$\mathrm{MSE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(I(i,j)-K(i,j))^2$$

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{MAX_I^2}{\mathrm{MSE}}\right)$$

SSIM

Structural Similarity Index

Perceptual metric comparing luminance, contrast, and structure between images.

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

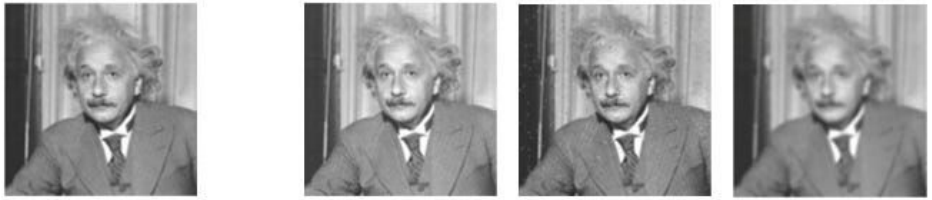
Entropy

Shannon Entropy

Measures information content.
Higher entropy => more detail recovered.

$$H = -\sum_{i=0}^{L-1} p(i)\log_2 p(i)$$

SNR: 40 dB SNR: 20 dB SNR: 10 dB SNR: 0 dB



Original SSIM=1 PSNR=26.547 SSIM=0.988 PSNR=26.547 SSIM=0.840 PSNR=26.547 SSIM=0.694

Tech Stack

Tool	Role	Purpose
`C++17`	**Core language**	All algorithms implemented from scratch — DFT, k-means, SSIM, PSNR, entropy
`CMake`	**Build system**	Compiles and links the project across platforms
`stb_image`	**Image I/O**	Load and save PNG/JPG files. Single header, zero dependencies
`Eigen`	**Matrix storage**	Stores pixel arrays only. No Eigen algorithms used
`OpenCV`	**Display only**	Shows images on screen.

Current Status

[COMPLETE]	***DFT / IDFT implementation***
[IN-PROGRESS]	K-Means ***segmentation***
[TODO]	***Per-region correction (blur, noise, exposure)***
[TODO]	***Evaluation*** (PSNR, SSIM, Entropy)
[TODO]	Optional: DCT/IDCT ***blockiness*** removal

References

Book	**Digital Image Processing**	Gonzalez & Woods — Ch. 4 (DFT), Ch. 8 (DCT/JPEG), Ch. 3 (Histograms), Ch. 10 (Segmentation)
Book	**Computer Vision: Algorithms and Applications**	Szeliski — Ch. 3: Image processing fundamentals
Paper	**SSIM — Wang et al. (2004)**	Image quality assessment: from error visibility to structural similarity. IEEE Trans. Image Processing.
Paper	**Pertuz et al. (2013)**	Analysis of focus measure operators for shape-from-focus. Pattern Recognition — covers frequency-based blur scores.
Paper	**MacQueen (1967)**	Some methods for classification and analysis of multivariate observations — original K-Means paper.